

LESSON 09 / 第九課

Tree Biomechanics under Extreme Wind: From the Wind Field to Residual Wood

極端風下的樹木生物力學：從風場到剩餘木材

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Extreme wind is a load case, not a moral event. At regional scale the climate supplies a velocity distribution and a saturation regime for root plates. At city scale, street canyons and upwind buildings rewrite the approach flow; Zhou and colleagues’ work after Typhoon Mangkhut is one published built-environment wind field. At the tree, drag scales approximately with the square of wind speed, damping dissipates energy, and thigmomorphogenesis has already allocated wood. At the section, residual geometry — not a t/R slogan — is what remains to carry moment. Bond and Gruber argued that t/R is not a stand-alone failure criterion. A post-storm inspection is a new assessment in a named envelope, not a ritual walk.

極端風是荷載工況，不是道德事件。區域尺度上，氣候提供速度分布與根盤飽和體制。城市尺度上，街道峽谷與上風建築改寫來流；Zhou 及同事在颱風山竹之後的工作，是已發表建成環境風場之一。樹上，阻力約與風速平方成正比，阻尼耗能，而觸變形態建成已分配過木材。截面上，剩餘幾何——不是一句 t/R 口號——才是剩餘去承擔彎矩的東西。Bond 與 Gruber 主張 t/R 不是獨立失效準則。災後檢查是在點名包絡中的新評估，不是儀式性散步。

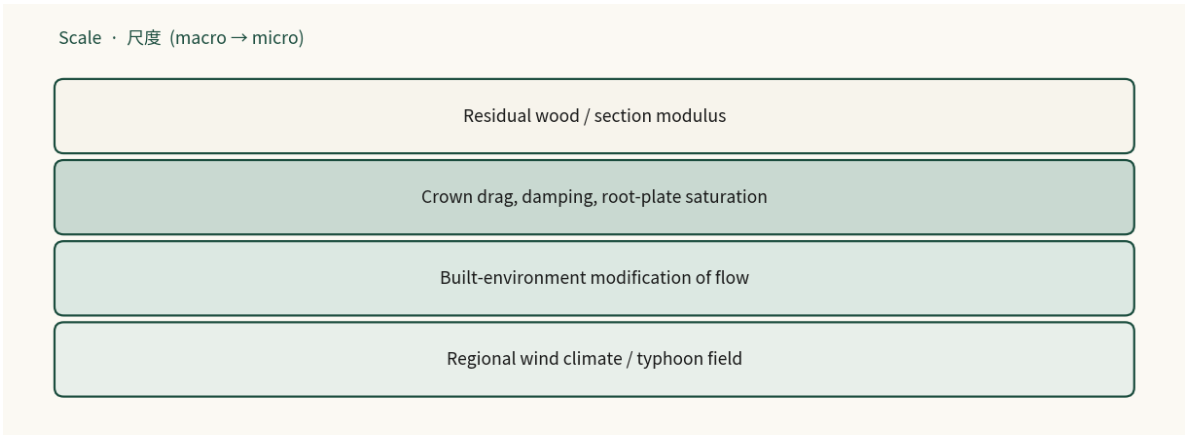


Figure 1. Nested scales from the regional wind field to residual wood.

圖 1. 由區域風場到剩餘木材的嵌套尺度。

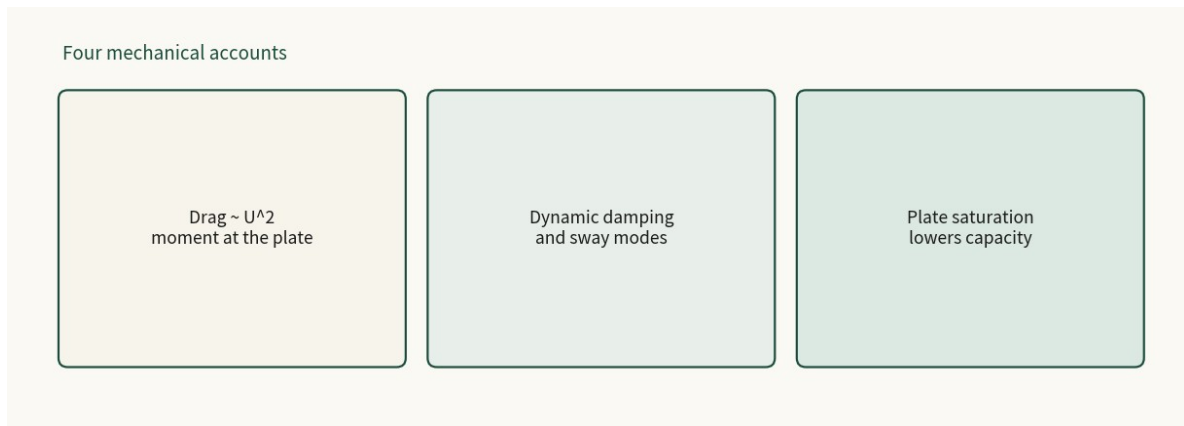


Figure 2. Drag, damping and plate saturation are different accounts.

圖2. 阻力、阻尼與盤飽和是不同賬戶。

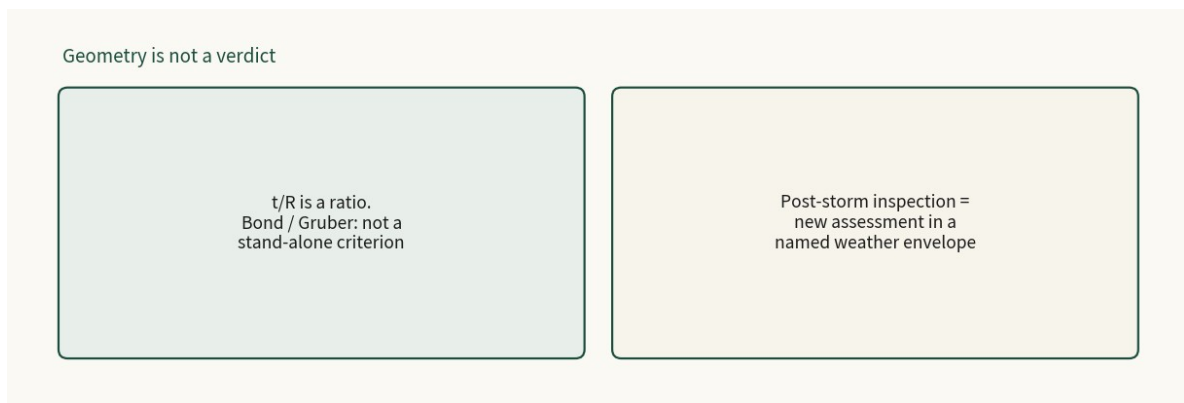


Figure 3. t/R informs; it does not close the sentence.

圖3. t/R 提供資訊；它不結束句子。

1. Scope

1. 範圍

The object is published wind-tree mechanics under extreme events, from the approach flow to residual section. A single typhoon name is an example of an envelope, not the frame.

對象是極端事件下已發表的風-樹力學：由來流到剩餘截面。單一颱風名是包絡的例子，不是框架。

2. Field, drag, damping

2. 風場、阻力、阻尼

Built form changes the flow that the crown actually sees. Zhou et al., working the Mangkhut damage field, treated urban wind as spatially structured rather than as a uniform free-stream [1]. Drag force on a porous crown scales approximately with U^2 ; moment at the root plate is therefore not linear in wind speed [2,3]. James and colleagues, and later sway studies, treat damping and modal behaviour as first-order properties, not as decorations on a static beam [4]. Thigmomorphogenesis allocates wood under repeated load; a tree grown in a sheltered canyon has a different wood account from one grown on an exposed ridge [5].

建成形態改變樹冠實際看見的流。Zhou 等處理山竹災損場時，把城市風當成空間結構的，而不是均勻自由流 [1]。多孔樹冠上的阻力約與 U^2 成正比；因此根盤彎矩對風速不是線性的 [2,3]。James

及同事以及後來的搖擺研究，把阻尼與模態當成一階屬性，而不是靜力梁上的裝飾 [4]。觸變形態建成在反覆荷載下分配木材；遮蔽峽谷中長大的樹，與暴露山脊上長大的樹，木材賬戶不同 [5]。

3. Plate saturation and residual wood

3. 盤飽和與剩餘木材

A saturated root plate loses capacity before the stem's residual wall is rewritten. Post-event soils that remain wet keep that reduced capacity in force. Residual-wall geometry informs the stem account. Bond and Gruber argued that treating t/R as a pass/fail law is not supported as a stand-alone criterion [6,7]. The operational sentence names the part, the weather envelope, and the residual geometry together.

飽和根盤在莖的剩餘壁被改寫之前就已失去容量。災後仍然潮濕的土壤使該降低容量繼續有效。剩餘壁幾何為莖賬戶提供資訊。Bond 與 Gruber 主張把 t/R 當成通過／失敗定律，並不作為獨立準則成立 [6,7]。操作句子把部位、天氣包絡與剩餘幾何一齊點名。

4. Limits

4. 限度

Most instrumented sway series are temperate. Tropical cyclone spectra, palm stems and stonewall-coupled trees are thinner in the same variables. A damping number copied from a plantation spruce is not a street-tree measurement.

多數儀器化搖擺序列是溫帶的。熱帶氣旋譜、棕櫚莖與石牆耦合樹在同一變量上較薄。從人工林雲杉抄來的阻尼數字，不是行道樹測量。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. Extreme wind is treated here as:

1. 此處把極端風視為：

- A. A moral event.
A. 道德事件。
- B. A load case.
B. 荷載工況。
- C. A mulch rule.
C. 覆蓋規則。
- D. Wall 4.
D. 第 4 壁。

2. Drag force scales approximately with:

2. 阻力大約正比於：

- A. Wind speed.
A. 風速。
- B. The square of wind speed.
B. 風速的平方。

C. Soil pH.

C. 土壤 pH。

D. Canopy percent only.

D. 只有樹冠百分比。

3. Built-environment wind is:

3. 建成環境風是：

A. Always a uniform free-stream.

A. 永遠是均勻自由流。

B. Spatially structured by canyons and buildings.

B. 被峽谷與建築空間結構化。

C. Equal to P50.

C. 等於 P50。

D. A register.

D. 一份名錄。

4. t/R in this review is:

4. 本綜述中的 t/R 是：

A. A stand-alone failure law.

A. 獨立失效定律。

B. Geometry that informs, not a stand-alone criterion.

B. 提供資訊的幾何，不是獨立準則。

C. A diversity rule.

C. 多樣性規則。

D. An inoculant.

D. 接種劑。

5. Plate saturation:

5. 盤飽和：

A. Raises capacity.

A. 提高容量。

B. Lowers plate capacity before the stem wall is rewritten.

B. 在莖壁被改寫前降低盤容量。

C. Equals Wall 1.

C. 等於第 1 壁。

D. Sets SOS.

D. 設定 SOS。

6. Thigmomorphogenesis:

6. 觸變形態建成：

A. Does not allocate wood.

A. 不分配木材。

B. Allocates wood under repeated load.

B. 在反覆荷載下分配木材。

C. Is a lightning path.

C. 是雷電路徑。

D. Is 3-30-300.

D. 是 3-30-300。

7. A post-storm walk is:

7. 災後散步是：

A. A ritual that closes risk.

A. 關閉風險的儀式。

B. A new assessment in a named envelope.

B. 在點名包絡中的新評估。

C. A P88 table.

C. P88 表。

D. A nursery system.

D. 苗圃系統。

8. Damping and modal behaviour are:

8. 阻尼與模態是：

A. Decorations on a static beam.

A. 靜力梁上的裝飾。

B. First-order mechanical properties.

B. 一階力學屬性。

C. Soil textures.

C. 土壤質地。

D. Cable colours.

D. 鋼纜顏色。

9. Moment at the plate versus wind speed is:

9. 盤上彎矩對風速是：

A. Linear.

A. 線性的。

B. Not linear, because drag scales $\sim U^2$.

B. 非線性，因為阻力約正比於 U^2 。

C. Zero always.

C. 總是零。

D. Equal to occupancy.

D. 等於佔用。

10. A damping number from plantation spruce used on a tropical street tree is:

10. 把人工林雲杉的阻尼數字用在熱帶行道樹上是：

A. A measurement of that street tree.

A. 該行道樹的測量。

B. A transfer, not that measurement.

B. 一項轉移，不是該測量。

C. Required by ANSI Part 4.

C. ANSI 第 4 部分所要求。

D. Wall 3.

D. 第 3 壁。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

- [1] Zhou et al. (2024). built-environment wind field and Typhoon Mangkhut damage.
- [2] Mayhead G.J. drag coefficients of trees.
- [3] Kane B., Smiley E.T. drag and failure of urban trees.
- [4] James K.R., Haritos N., Ades P.K. mechanical stability of trees under dynamic loads.
- [5] Telewski F.W. thigmomorphogenesis reviews.
- [6] Bond J. (2006). t/R critique.
- [7] Gruber F. (2008). t/R cannot predict stem failure.
- [8] Peltola H. windthrow mechanics in stands and on exposed trees.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。